Practice Problems
Calculus Physics II

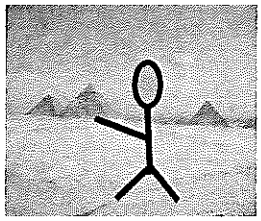
**Name:** Key **Date:** _____

## Chapter 62 Optical Instruments

1) Angular magnification is

A) The ratio of the image height to the object height.
B) The ratio of the image distance to the object distance.
C) The amount of the retina that is covered by the image.
D) How large an image is.

2) Cathy and Mike are in Egypt and for fun Cathy takes a picture of Mike with his hand out and it looks like he is holding in his hand one on the Pyramid's, why does the Pyramid and Mike's hand look the same sizes?



A) They are the same size.
B) The angular magnification of the Pyramid is larger than Mike's hand.
C) The angular magnification of Mike's hand is larger than the Pyramid.
D) The image size on the retina is the same size.

3) Mike is looking at a tree with the unaided eye and the tree make an angle of 0.25° with his eye, he uses a telescope and the angle with his eye changes to 1.5°, what is the angular magnification of the telescope?

$$M = \frac{\theta_{aid}}{\theta_{unaid}} = \frac{1.5^\circ}{0.25^\circ} = 6$$

4) To increase the magnification of a magnifying glass you should use a lens with a

A) larger focal length
B) smaller focal length
C) larger diameter
D) smaller diameter

$$m = \frac{25\text{ cm}}{f}$$

larger $f$ → Larger $m$

5) A convergent lens with a focal length of the lens is 5 cm, is held above a road on the map, what is the angular magnification of the road on the map?

$$m = \frac{25\text{cm}}{f} = \frac{25\text{ cm}}{5\text{ cm}} = 5$$

6) A convergent lens is used as a magnifying glass, if a virtual image is formed at your near point N = 25 cm and the angular magnification is M = 9, what is the focal length of the lens?

$$m = \frac{25\text{cm}}{f} \quad | \quad f = \frac{25\text{cm}}{9} = 2.78\text{ cm}$$

7) A penny is placed 2.5 cm from a magnifying glass with a focal length of 3.0 cm, what is the linear magnification (m) and what is the angular magnification (M) of the image of the penny?

Linear

$d_o = 2.5\text{cm}$

$f = 3.0\text{cm}$

$$d_i = \frac{1}{\frac{1}{f} - \frac{1}{d_o}} = -15\text{cm}$$

$$m = -\frac{d_i}{d_o}$$

$m = 6$

angular

$$m = \frac{25\text{cm}}{f} = 8.3$$

8) A microscope has an objective lens of focal length 0.140 cm and an eyepiece of focal length 2.0 cm. It is adjusted for minimum eyestrain for persons with a near point of 25.0 cm. A blood sample is placed 0.15 cm from the objective. The angular magnitude of the microscope is closest to

$f_o = 0.14\text{cm}$
$f_e = 2\text{ cm}$
$d_o = 0.15\text{cm}$

need $d_i$ using $f_o$

$$d_i = \frac{1}{\frac{1}{f_o} - \frac{1}{d_o}} = 2.1\text{cm}$$

$$M = -\frac{d_i N}{f_o f_e} = -\frac{(2.1\text{cm})(25\text{cm})}{(0.14\text{cm})(2\text{cm})}$$

$$M = -1875$$

9) A microscope consists of two convergent lenses that are 30 cm apart, the objective has a focal length of $f_o = 1.5$ cm, the eye piece has a focal length of $f_e = 25$ cm, if the object is placed 1.6 cm from the objective lens, what is the angular magnification of the microscope? (Assume the near point N = 25 cm)

$f_o = 1.5\text{cm}$
$f_e = 25\text{cm}$
$d_o = 1.6\text{cm}$

$$d_i = \frac{1}{\frac{1}{f} - \frac{1}{d_o}} = 2.5\text{cm}$$

$$M = -\frac{d_i N}{f_o f_e} = -16$$

10) The objective lens of an astronomical telescope has a focal length of 60 cm and the eyepiece has a focal length of 2 cm, what is the magnification of the telescope?

$$M = -\frac{f_o}{f_e} = -\frac{60\text{cm}}{2\text{cm}} = -30$$

11) The focal length of the objective of a telescope is 120 cm. The angular magnifying of this telescope is M = - 20, what is the focal length of the eyepiece of the of the telescope?

$f_o = 120\text{cm}$
$M = 20$

$$M = -\frac{f_o}{f_e}$$

$$f_e = \frac{f_o}{m} = 6\text{ cm}$$

12) If Bob wants just Cathy in focus and the background blurry and out of focus in a picture, what does Bob do to the f-number.

A) Uses a large f-number
(B) Uses a small f-number
C) f-number does not affect the focus

13) The focal length of a thin lens is 40 mm and its aperture diameter is 10 mm. What is the *f*-number of this lens?

$$\# = \frac{f}{D} = \frac{40\text{ mm}}{10\text{ mm}} = 4$$

14) The "Depth of field" for a camera is given by the f/#, if the camera is set to f/22 and the single lens camera has a focal length of 30 cm, what must be the diameter of the aperture to achieve this f-number?

$$\# = 22 = \frac{f}{D}$$

$$D = \frac{f}{\#} = \frac{30\text{cm}}{22} = 1.36\text{cm}$$

15) The focal length of the lens of a simple camera is 2.0 cm. How far must the lens be moved to change the focus of the camera from a person 25 m away to one that is 0.5 m away?

$d_{o_1} = 25\,m$

$f = 2\,cm = 2\times10^{-2}\,m$

$d_{i_1} = \frac{1}{\frac{1}{f} - \frac{1}{d_{o_1}}} = 0.02000\,cm$ or $2.000\,cm$

$d_{o_2} = 0.5\,m$

$f = 2\times10^{-2}\,m$

$d_{i_2} = \frac{1}{\frac{1}{f} - \frac{1}{d_{o_2}}} = 0.02083\,m$ or $2.083\,cm$

$d_{i_2} - d_{i_1} = 0.00083\,m$ or $0.83\,mm$

16) Passive Auto Focusing uses

(A) The contrast of the image on the CCD.
B) Either sound or light to gauge the distance and then sets the focus.
C) The CCD to gauge the distance and then sets the focus.

17) What type of lens do light house use to collimate the light.

A) Convergent lenses
B) Divergent lenses
C) Meniscus lenses
(D) Fresnel lenses

18) In a fiber optic cable how does the index of refraction of the core compare to the index refraction of the cladding material?

(A) Index of refraction of core must be larger than the index of refraction of the cladding.
B) Index of refraction of core must be smaller than the index of refraction of the cladding.
C) Index of refraction of core must be the same as the index of refraction of the cladding.

19) A plastic fiber optic cable has a core made of acrylic (n = 1.49) and a cladding of a fluoropolymer (n = 1.36), what is the minimum angle that light will experience total internal reflection in the fiber core?

$\theta_c = \sin^{-1}\left(\frac{n_{clad}}{n_{core}}\right) = \sin^{-1}\left(\frac{1.36}{1.46}\right) = 65.9^0$

20) A sign seems to glow at night when the car head lights shine on it, what type of reflector is in the sign?

A) Elliptical reflector
B) Parabolic reflector
C) Spherical reflector
(D) Retro Reflector

21) Which reflector will focus light at the second foci when the light is placed at the first foci?

A) Elliptical reflector
B) Parabolic reflector
C) Spherical reflector
D) Retro Reflector

22) Why can you see heat in a Schlieren imaging system?

A) Because of the different index of refraction of the hot air compared to cooler air.
B) Because of hot air has the same index of refraction as the cooler air.
C) Because the dispersion of the hot air is different than cooler air.
D) Because the scattering of light in hot air is different than cooler air.

KEY

1) C
2) D
3) 6
4) B
5) 5
6) 2.78 cm
7) 8.3
8) – 187.5
9) – 16
10) – 30
11) 6 cm
12) B
13) 4
14) 1.36 cm
15) 0.083 cm
16) A
17) D
18) A
19) 65.9°
20) D
21) A